KaroSpace: a rapid-access framework for interactive exploration of multi-sample spatial omics data



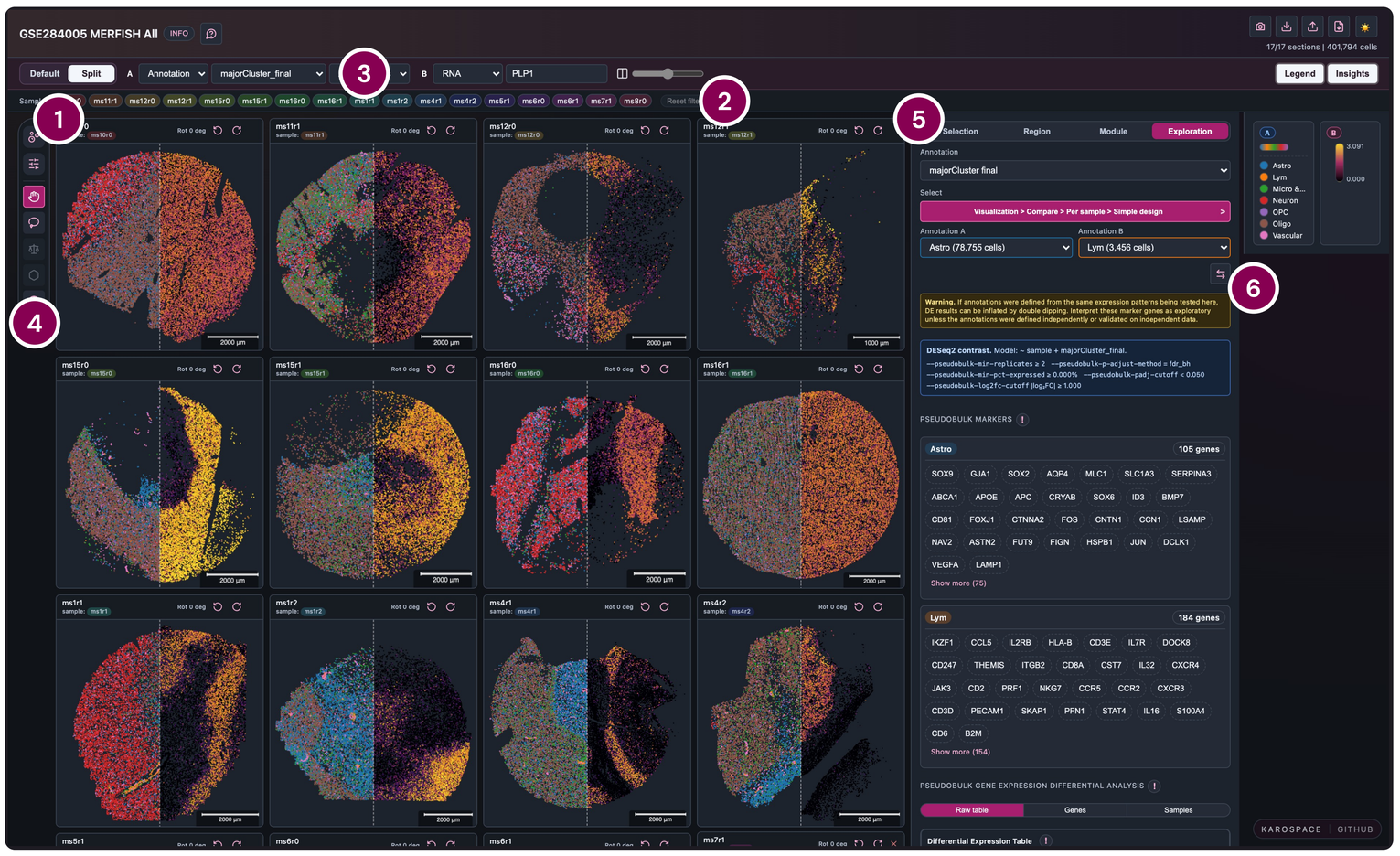
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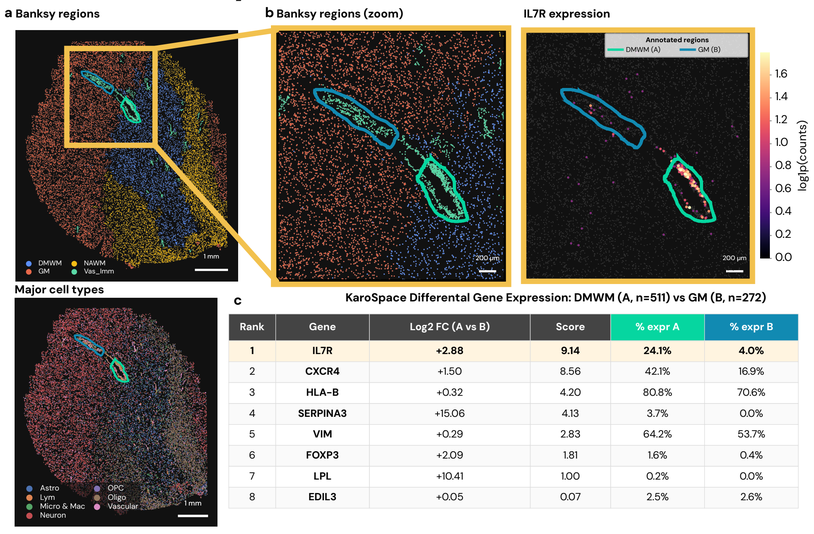
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50+	10	5	1
public viewers	platforms	modalities	HTML file

a One viewer for many samples · 17 MERFISH sections, 402k cells, sample-level DESeq2



c Case study: a missed niche



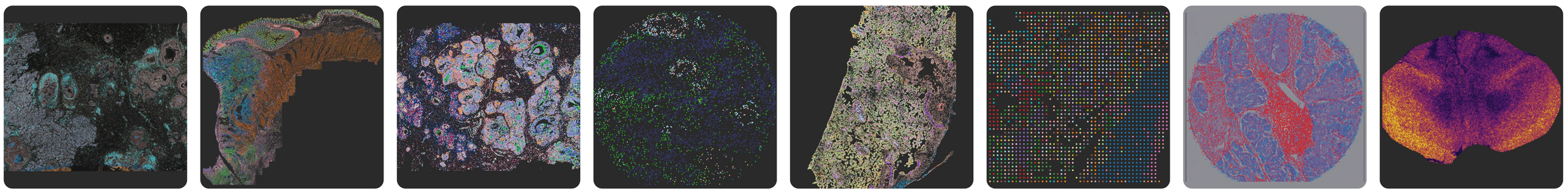
c KaroSpace Differential Gene Expression: DMWM (A, n=511) vs GM (B, n=272)

Rank	Gene	Log2 FC (A vs B)	Score	% expr A	% expr B
1	**IL7R**	**+2.88**	**9.14**	**24.1%**	**4.0%**
2	CXCR4	+1.50	8.56	42.1%	16.9%
3	HLA-B	+0.32	4.20	80.8%	70.6%
4	SERPINA3	+15.06	4.13	3.7%	0.0%
5	VIM	+0.29	2.83	64.2%	53.7%
6	FOXP3	+2.09	1.81	1.6%	0.4%
7	LPL	+10.41	1.00	0.2%	0.0%
8	EDIL3	+0.05	0.07	2.5%	2.6%

Banksy compartments → draw polygons
DMWM (A) vs GM (B) on one lesion section, then run region-vs-region DE in the viewer.

IL7R⁺ vascular inflammatory niche
IL7R top hit (log2FC +2.88; 24% vs 4% cells), with CXCR4, FOXP3: a niche beyond the original study (Feng et al., 2025).

b Same viewer, any platform or modality



Xenium + H&E	CosMx	Atera	Visium HD	Stereo-seq	DBiT-seq	COMET + Xenium	MALDI-MSI
RNA · image overlay	RNA · colon	RNA · breast cancer	RNA · binned	RNA · whole section	RNA · mouse embryo	protein · 16-plex	lipidomics · brain

Fig. 1 · *(a) MERFISH data (Feng et al., 2025): cell types | PLP1 in split view; numbered callouts match Key features (3). (b) Public datasets across platforms, each a single exported viewer. (c) Banksy compartments and in-browser region DE resolve a distinct vascular inflammatory (IL7R⁺) niche beyond the original study.*

1 · Motivation

Spatial omics resolves molecular & cellular organization — yet exploration is bottlenecked by proprietary, compute-heavy, tightly-coupled tools.

KaroSpace: lightweight, pipeline-agnostic, cell-centric — portable & reproducible.

2 · Pipeline

Spatial features — H5AD · .zarr · .rds
↓
Companion · Rust — precompute metrics
↓
KaroSpace · Python — reuse & render
↓
Viewer — HTML · sidecar · .karospace

3 · Key features

1. Multi-sample grid, zoom/pan
2. Metadata-based sample filters
3. Colour by annotation, gene, module
4. Lasso, polygon regions, cell query
5. Insights: markers, enrichment, DESeq2
6. Legend spotlight, editable

- ✓ Tissue image overlays (DAPI, H&E)
- ✓ Linked UMAP ↔ section lasso
- ✓ Region-to-region comparison
- ✓ Split view with order slider
- ✓ Sessions & regions export (JSON)
- ✓ Light/dark, screenshots
- ✓ Embedded story-mode tutorial

4 · Datasets & platforms

Platforms: Visium/HD, Xenium, HybISS, MERFISH, CosMx, OpenST, Array-Seq, DBiT-Seq, Atera.

Modalities: transcriptomics, epigenomics, proteomics, lipidomics, multiome.

5 · Analytics & statistics

Cell: Welch t-test, log2FC, %expr (BH).

Sample: replicate-aware DESeq2.

User-defined region DE — see Fig. 1c.

6 · Availability

web · code · agent · paper

Reference:
1. Feng, R. *et al.* Single-cell spatial transcriptomic profiling defines a pathogenic inflammatory niche in chronic active multiple sclerosis lesions. Immunity 58, 2989-3005.e10 (2025)
2. Langseth, C. M. *et al.* KaroSpace: a rapid-access framework for interactive exploration of multi-sample spatial omics data. https://doi.org/10.64898/2026.03.10.710790 (2026)

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